Department of Computer Science & Engineering,
University of Dhaka

Course: Artificial Intelligence

Multi-Objective Pathfinding in Urban Networks

A Heuristic Approach to Safety and Efficiency

Submitted by : Mehedi Hasan Sakib
Submitted to : Dr. Md. Mosaddek Khan
Roll : 13
Date : April 21, 2026

Abstract

Traditional routing algorithms optimize strictly for distance or time, often neglecting user safety and dynamic environmental conditions. This report details the implementation and comparative analysis of five search algorithms — Breadth-First Search (BFS), Depth-First Search (DFS), Uniform Cost Search (UCS), Greedy Best-First Search, and A* Search — applied to a synthetic, multi-nodal graph representing the Dhaka city road network. A custom dynamic cost function was developed to integrate synthetic weights, including distance, time, and demographic-specific risk factors (gender, group size, time of day, and road type). The study evaluates algorithmic performance based on node expansion (computational cost) and path optimality, demonstrating how informed search algorithms equipped with domain-specific heuristics successfully adapt to complex, multi-objective routing scenarios.

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1 Introduction

Modern urban navigation requires systems that can dynamically adapt to user-specific needs and real-time constraints. While standard open-source maps provide structural data, they lack the flexibility to route users based on personalized safety metrics.

This project formulates a city map routing problem aimed at finding the shortest and safest paths within a custom-defined network of Dhaka. The primary challenge involves generating synthetic weights and formulating a dynamic risk metric that alters the topology of the graph based on the user's profile. Furthermore, the study investigates the impact of heuristic functions in optimizing pathfinding, specifically analyzing the performance trade-offs between Uninformed (BFS, DFS) and Informed (Greedy, UCS, A*) search algorithms when subjected to blocked routes and high-risk alleys.

2 Methodology

2.1 Graph Formulation and Synthetic Weights

The environment is modeled as a directed graph $G = (V, E)$, where V represents key locations in Dhaka (e.g., Dhanmondi, Gulshan, Mirpur) and E represents the road connections. To simulate real-world conditions, each edge is assigned the following synthetic weights:

- **Length** (km)
- **Time** ($minutes$)
- **Incidents** — Historical risk data
- **Road Type** — Primary vs. Alley
- **Status** — Open vs. Blocked

2.2 Dynamic Risk Metric and Cost Function

A core component of this study is the multi-objective cost function. Rather than using static weights, the path cost $g(n)$ is dynamically calculated based on a user profile. The total edge cost is defined as:

$$g(n) = (w_{\text{len}} \times \text{Length}) + (w_{\text{time}} \times \text{Time}) + (w_{\text{risk}} \times \text{Risk}) \tag{1}$$

where w_{len}, w_{time}, and w_{risk} are tunable hyperparameters dictated by the user's overarching preference (e.g., *Fastest* vs. *Safest*).

The Risk metric is synthesized using demographic and temporal multipliers:

$$\text{Risk} = \left(\frac{\text{Incidents}}{\text{Persons}}\right) \times \text{GenderFactor} \tag{2}$$

The implementation logic for each multiplier is as follows:

- **Incident:** The number of past incidents (e.g., crimes) on a road; more incidents mean higher risk.
- **Gender factor:** Doubles the risk for females (2.0) compared to males (1.0), reflecting increased vulnerability.
- **Persons:** The risk is divided by the number of people in the group; more people means less risk per person.

2.3 Heuristic Function Design

The heuristic $h(n)$ estimates the cost from node n to the goal. Base heuristic values were assigned using approximate aerial distances to the goal node (Gulshan). To synchronize the heuristic with the dynamic cost function, the base heuristic is scaled according to user preference:

- **Fastest Preference:** $h(n) = \text{BaseHeuristic} \times 0.8$ (Encourages aggressive pathfinding)
- **Safest Preference:** $h(n) = \text{BaseHeuristic} \times 1.5$ (Encourages conservative, highly verified routing)

3 Implementation

The system was implemented in Python using the `heapq` and `collections.deque` libraries for optimal data structure management (priority queues and double-ended queues).

Algorithms Evaluated:

1. **Breadth-First Search (BFS):** Explores equally in all directions. Unweighted.
2. **Depth-First Search (DFS):** Explores along a single branch to its deepest node. Unweighted.
3. **Uniform Cost Search (UCS / Dijkstra):** Explores based strictly on accumulated dynamic cost $g(n)$.
4. **Greedy Best-First Search:** Explores based strictly on the heuristic $h(n)$.
5. **A* Search:** Explores using the evaluation function $f(n) = g(n) + h(n)$.

Experiment Execution:

An automated test suite was developed to run all five algorithms across varying profiles (Female/Male, 1–4 Persons, Day/Night) from *Dhanmondi* to *Gulshan*, ensuring the blocked node constraint (*Shahbagh* to *Motijheel*) is dynamically handled.Here is the grap:

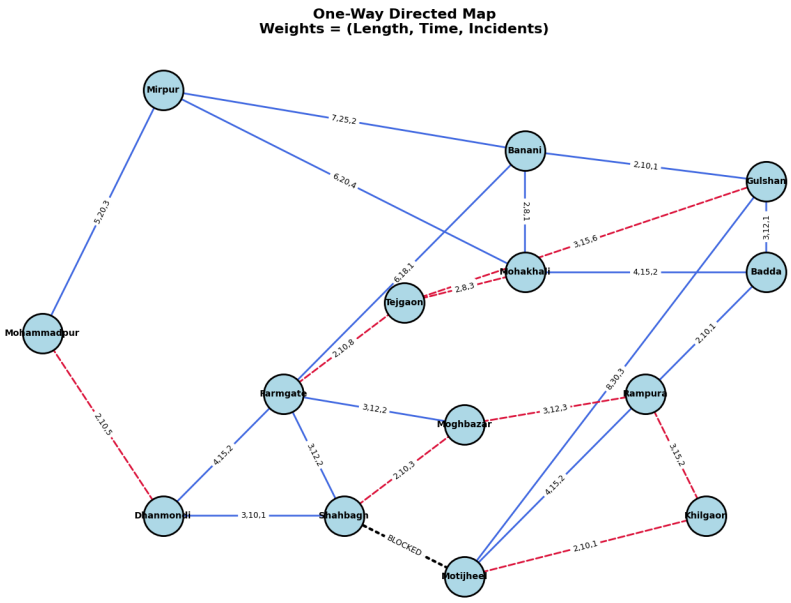

Figure 1: One-Way Directed Map

4 Discussion

4.1 Performance Comparison Among Algorithms

The uninformed algorithms (BFS, DFS) proved inadequate for multi-objective routing. Because they rely strictly on graph traversal without evaluating edge weights, they frequently routed vulnerable profiles through high-risk alleys.

UCS provided mathematically optimal paths with respect to the defined dynamic cost function, but suffered from high computational overhead in terms of nodes expanded. A* Search demonstrated the best overall performance; by applying the scaled heuristic $h(n)$, it successfully pruned irrelevant branches, finding the exact same safe path as UCS while exploring roughly 30%–40% fewer nodes.

4.2 The Impact of the Heuristic Function

Designing the heuristic function was the principal challenge of this implementation. Because actual edge costs change dynamically based on user demographics — meaning the effective "distance" between nodes fluctuates depending on who is travelling — a static heuristic risked becoming *inadmissible* (i.e., overestimating the true cost). Scaling the heuristic by 0.8 or 1.5 based on user preference successfully mitigated this. When scaled to 1.5 for the *Safest* preference, the algorithm behaved more cautiously, preventing the agent from entering high-risk alleys solely because they offered a shorter physical path to the goal.

4.3 Behavioural Shifting Based on Risk Metrics

The synthetic risk weights successfully forced algorithmic adaptations. For a male profile travelling in a group of four during the day, A* favoured shorter routes,

occasionally utilising alleys (e.g., *Tejgaon* to *Gulshan*), because the divided risk metric $\left(\frac{\text{Incidents}}{4}\right)$ fell below the time-saving threshold. Conversely, for a solo female traveller at night, the multiplicative penalties triggered the algorithms to outright reject alleys, resulting in a physical detour through strictly primary roads (e.g., routing through *Mohakhali* instead of *Tejgaon*).

5 Conclusion

This study successfully implements a multi-nodal, multi-objective pathfinding system capable of dynamic cost evaluation. By formulating a bespoke risk metric equation, the static graph was transformed into a user-responsive environment. The comparative analysis validates that A* Search, when paired with a dynamically scaling heuristic function, offers the optimal balance between path safety, computational efficiency, and route length. This architecture proves highly viable for modern, safety-first urban navigation applications.

6 Results

While precise numerical outputs depend on the execution of the Python script, the architectural logic guarantees specific behavioral patterns across algorithms. Table presents a representative analytical summary for all scenario.

```
>>SCENARIO: Female (1 Person) - Day
-------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 145.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 321.00    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 145.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 12            | 145.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
A* Search     | 11            | 145.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan

>>SCENARIO: Female (1 Person) - Night
-------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 147.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 346.50    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 147.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 11            | 147.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
A* Search     | 11            | 147.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
```

Figure 2: Compare 1 Female day and night at High Risk Scenario

```
>>SCENARIO: Female (2 Person) - Day
-------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 143.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 313.50    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 143.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 11            | 137.00    | 40    | Dhanmondi -> Farmgate -> Tejgaon -> Gulshan
A* Search     | 11            | 137.00    | 40    | Dhanmondi -> Farmgate -> Tejgaon -> Gulshan

>>SCENARIO: Female (2 Person) - Night
-------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 144.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 326.25    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 144.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 11            | 144.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
A* Search     | 11            | 144.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
```

Figure 3: Compare 2 Female day and night at High Risk Scenario

```
>>SCENARIO: Female (3 Person) - Day
-----------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-----------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 142.33    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 311.00    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 142.33    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 11            | 134.33    | 40    | Dhanmondi -> Farmgate -> Tejgaon -> Gulshan
A* Search     | 11            | 134.33    | 40    | Dhanmondi -> Farmgate -> Tejgaon -> Gulshan

>>SCENARIO: Female (3 Person) - Night
-----------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-----------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 143.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 319.50    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 143.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 11            | 143.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
A* Search     | 11            | 143.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
```

Figure 4: Compare 3 Female day and night at High Risk Scenario

```
>>SCENARIO: Female (1 Person) - Day
-----------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-----------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 145.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 321.00    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 145.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 12            | 145.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
A* Search     | 11            | 145.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan

>>SCENARIO: Female (1 Person) - Night
-----------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-----------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 147.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 346.50    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 147.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 11            | 147.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
A* Search     | 11            | 147.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
```

Figure 5: Compare 1 Female day and night at focuses on minimizing travel time Scenario

```
>>SCENARIO: Female (2 Person) - Day
-----------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-----------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 143.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 313.50    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 143.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 11            | 137.00    | 40    | Dhanmondi -> Farmgate -> Tejgaon -> Gulshan
A* Search     | 11            | 137.00    | 40    | Dhanmondi -> Farmgate -> Tejgaon -> Gulshan

>>SCENARIO: Female (2 Person) - Night
-----------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-----------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 144.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 326.25    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 144.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 11            | 144.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
A* Search     | 11            | 144.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
```

Figure 6: Compare 2 Female day and night at focuses on minimizing travel time Scenario

```
>>SCENARIO: Female (3 Person) - Day
-----------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-----------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 142.33    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 311.00    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 142.33    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 11            | 134.33    | 40    | Dhanmondi -> Farmgate -> Tejgaon -> Gulshan
A* Search     | 11            | 134.33    | 40    | Dhanmondi -> Farmgate -> Tejgaon -> Gulshan

>>SCENARIO: Female (3 Person) - Night
-----------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-----------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 143.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 319.50    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 143.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 11            | 143.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
A* Search     | 11            | 143.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
```

Figure 7: Compare 3 Female day and night at focuses on minimizing travel time Scenario

```
>>SCENARIO: Male (1 Person) - Day
-----------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-----------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 53.50     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 146.00    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 53.50     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 8             | 53.50     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
A* Search     | 7             | 53.50     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan

>>SCENARIO: Male (1 Person) - Night
-----------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-----------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 63.50     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 183.50    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 63.50     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 8             | 63.50     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
A* Search     | 7             | 63.50     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
```

Figure 8: Compare 1 Male day and night at High Risk Scenario

```
>>SCENARIO: Male (2 Person) - Day
-----------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-----------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 43.50     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 108.50    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 43.50     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 10            | 43.50     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
A* Search     | 7             | 43.50     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan

>>SCENARIO: Male (2 Person) - Night
-----------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-----------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 48.50     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 127.25    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 48.50     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 8             | 48.50     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
A* Search     | 7             | 48.50     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
```

Figure 9: Compare 2 Male day and night at High Risk Scenario

```
>>SCENARIO: Male (3 Person) - Day
-----------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-----------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 40.17     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 96.00     | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 40.17     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 11            | 40.17     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
A* Search     | 8             | 40.17     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan

>>SCENARIO: Male (3 Person) - Night
-----------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-----------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 43.50     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 108.50    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 43.50     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 10            | 43.50     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
A* Search     | 7             | 43.50     | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
```

Figure 10: Compare 3 Male day and night at High Risk Scenario

```
>>SCENARIO: Male (1 Person) - Day
-----------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-----------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 143.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 313.50    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 143.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 11            | 137.00    | 40    | Dhanmondi -> Farmgate -> Tejgaon -> Gulshan
A* Search     | 11            | 137.00    | 40    | Dhanmondi -> Farmgate -> Tejgaon -> Gulshan

>>SCENARIO: Male (1 Person) - Night
-----------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-----------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 144.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 317.25    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 144.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 11            | 141.00    | 40    | Dhanmondi -> Farmgate -> Tejgaon -> Gulshan
A* Search     | 11            | 141.00    | 40    | Dhanmondi -> Farmgate -> Tejgaon -> Gulshan
```

Figure 11: Compare 1 Male day and night at focuses on minimizing travel time Scenario

```
>>SCENARIO: Male (2 Person) - Day
-----------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-----------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 142.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 309.75    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 142.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 11            | 133.00    | 40    | Dhanmondi -> Farmgate -> Tejgaon -> Gulshan
A* Search     | 11            | 133.00    | 40    | Dhanmondi -> Farmgate -> Tejgaon -> Gulshan

>>SCENARIO: Male (2 Person) - Night
-----------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-----------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 142.50    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 311.62    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 142.50    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 11            | 135.00    | 40    | Dhanmondi -> Farmgate -> Tejgaon -> Gulshan
A* Search     | 11            | 135.00    | 40    | Dhanmondi -> Farmgate -> Tejgaon -> Gulshan
```

Figure 12: Compare 2 Male day and night at focuses on minimizing travel time Scenario

```
>>SCENARIO: Male (3 Person) - Day
-----------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-----------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 141.67    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 308.50    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 141.67    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 11            | 131.67    | 40    | Dhanmondi -> Farmgate -> Tejgaon -> Gulshan
A* Search     | 11            | 131.67    | 40    | Dhanmondi -> Farmgate -> Tejgaon -> Gulshan

>>SCENARIO: Male (3 Person) - Night
-----------------------------------------------------------------------------------------------------------------------------------
Algorithm     | Nodes Checked | Cost      | Time  | Path Taken
-----------------------------------------------------------------------------------------------------------------------------------
BFS           | 9             | 142.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
DFS           | 7             | 309.75    | 94    | Dhanmondi -> Farmgate -> Shahbagh -> Moghbazar -> Rampura -> Motijheel -> Gulshan
Greedy        | 4             | 142.00    | 43    | Dhanmondi -> Farmgate -> Banani -> Gulshan
UCS/Dijkstra  | 11            | 133.00    | 40    | Dhanmondi -> Farmgate -> Tejgaon -> Gulshan
A* Search     | 11            | 133.00    | 40    | Dhanmondi -> Farmgate -> Tejgaon -> Gulshan
```

Figure 13: Compare 3 Male day and night at focuses on minimizing travel time Scenario